Module 03d — OOP Part 4: Abstract Classes, Abstract Methods & Interfaces

Phase: Advanced Python
Previous module: OOP Part 3 — Polymorphism & Project
Next module: Exception Handling

Series: OOP overview · Part 1 · Part 2 · Part 3 · Part 4 (this file)

Lab kit: 15_Lab_Assets/32_campus_wallet_abc/

Module Outcome:

- Abstract method
- Abstract class (`ABC` , when we can / cannot construct)
- Interface in Python (ABC with only abstract methods)
- Concrete class vs abstract class vs interface
- SecureBank campus wallet (lab)

1. Abstract method

Real-world analogy: A **job posting** says “must treat patients.” It does not write the treatment. Each doctor fills that in. An **abstract method** is the must-have skill with **no recipe yet**. **SecureBank** declares `kind()` before savings or wallet exists. **AcmeHR** declares `net()` before intern vs faculty. **TechMart** declares `pay()` before UPI vs card.



Where used? SecureBank `kind()` , **AcmeHR** `net()` , **TechMart** `pay()` — the parent names the method; the child writes the body.

Definition: Sometimes we don't know the implementation yet, but we still want to declare the method. Such a method is an **abstract method**. It has only a **declaration**, not the real code.

Purpose: The parent says “every child must have this method.” The child writes the actual body.

```
from abc import ABC, abstractmethod


@abstractmethod
def m1(self):
    pass
```

`pass` means “no algorithm yet.” `@abstractmethod` lives in the `abc` module. Forget the import → `NameError` .

Case — forgot the import:

```
class Test:
    @abstractmethod
```

```
    def m1(self):
        pass
```

Expected Output:

```
NameError: name 'abstractmethod' is not defined
```

Child classes are responsible for providing the implementation.

Remember: Abstract method = “I know the name. I do not know the body yet.”

Write an AcmeHR abstract `JobRole` with `net_pay()`. Implement it on `Intern` and print `Intern().net_pay()`.

```
from abc import ABC, abstractmethod


class JobRole(ABC):
    @abstractmethod
    def net_pay(self):
        pass


class Intern(JobRole):
    def net_pay(self):
        return 15000


print(Intern().net_pay())
```

Expected Output:

```
15000
```

Write a SecureBank abstract `Account` with `kind()`. Implement it on `SavingsAccount` and `WalletAccount` and print both kinds.

```
from abc import ABC, abstractmethod


class Account(ABC):
    def __init__(self, holder):
        self.holder = holder

    @abstractmethod
    def kind(self):
        pass


class SavingsAccount(Account):
    def kind(self):
        return "savings"


class WalletAccount(Account):
```

```
    def kind(self):
        return "wallet"


print(SavingsAccount("Priya").kind())
print(WalletAccount("Ravi").kind())
```

Expected Output:

```
savings
wallet
```

---

2. Abstract class

Real-world analogy: An **incomplete house plan**. Rooms are named; the kitchen is not drawn. You cannot live in the plan — only in a finished house. Shared plumbing stays on the plan (`deposit`, holder). Children finish the missing room (`kind()`, `title()`).



Where used? SecureBank `Account` (savings vs wallet), **MedCare** `Report` (lab vs invoice).

Definition: Sometimes the implementation of a class is **not complete**. Such a partially implemented class is an **abstract class**.

Purpose: Share the parts we already know and force children to finish the missing part.

Every abstract class in Python should extend `ABC`. Instantiation depends on **two** things together: did we extend `ABC`, and is there at least one `@abstractmethod`?

Case-1 — Normal class, no abstract method

```
from abc import ABC, abstractmethod


class Test:
    pass


t = Test()
print("object created")
```

Expected Output:

```
object created
```

Concrete class — we can create an object.

Case-2 — Extends `ABC`, but no abstract method

```
from abc import ABC, abstractmethod
```

```
class Test(ABC):
    pass


t = Test()
print("object created")
```

Expected Output:

```
object created
```

`ABC` alone is not enough. No abstract method → Python treats it as complete.

Case-3 — Extends `ABC` and has an abstract method

```
from abc import ABC, abstractmethod


class Test(ABC):
    @abstractmethod
    def m1(self):
        pass


t = Test()
```

Expected Output:

```
TypeError: Can't instantiate abstract class Test ... m1
```

(The exact wording changes slightly by Python version.)

Case-4 — Abstract method, but does not extend `ABC`

```
from abc import ABC, abstractmethod


class Test:
    @abstractmethod
    def m1(self):
        pass


t = Test()
print("object created")
```

Expected Output:

```
object created
```

Without `ABC`, the decorator is only documentation. It does not block `Test()`.

Case-5 — Abstract method with a real body, still not extending `ABC`

```
from abc import ABC, abstractmethod


class Test:
    @abstractmethod
    def m1(self):
        print("Hello")


t = Test()
t.m1()
```

Expected Output:

```
Hello
```

Without `ABC`, the method runs like a normal method.

Conclusion: At least one abstract method **and** `ABC` → instantiation is not possible.

Remember: abstract class + abstract method → no object.

A child that skips the abstract method is **also abstract**. The `class` line is legal. `Child()` is not.

```
from abc import ABC, abstractmethod


class Vehicle(ABC):
    @abstractmethod
    def no_of_wheels(self):
        pass


class Bus(Vehicle):
    pass


print("Bus class exists. No object yet — no error.")
# b = Bus()  # TypeError — Bus did not override no_of_wheels
```

Expected Output:

```
Bus class exists. No object yet — no error.
```

When children **do** implement the method:

```
from abc import ABC, abstractmethod


class Vehicle(ABC):
    @abstractmethod
    def no_of_wheels(self):
        pass
```

```
class Bus(Vehicle):
    def no_of_wheels(self):
        return 6

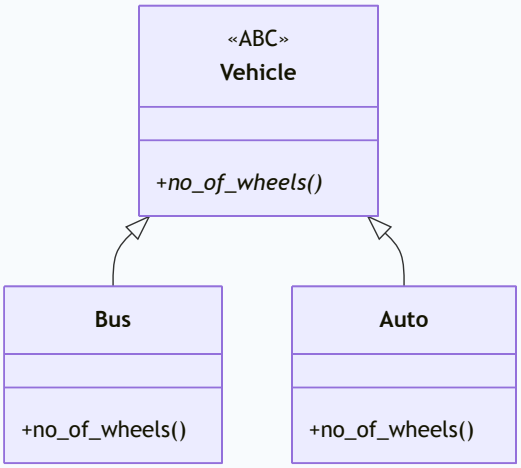

class Auto(Vehicle):
    def no_of_wheels(self):
        return 3


print(Bus().no_of_wheels())
print(Auto().no_of_wheels())
```

Expected Output:

```
6
3
```

An abstract class can contain **both** abstract and normal methods. Normal methods are the shared implementation. Abstract methods are the holes.



Key takeaways:

- `ABC` alone does not block objects
- `@abstractmethod` alone does not block objects
- **Both together** block objects
- A child that skips the abstract method is still abstract

Interview: Can an abstract class have a constructor? **Yes.** `__init__` runs when we construct the **child**, via `super()`. We never construct the abstract parent.

Common Error: `class Account(ABC):` with no `@abstractmethod`. Then `Account("Campus")` still works — Case-2.

Write a SecureBank abstract `Account` with shared `deposit` / `get_balance` and abstract `kind()`. Construct `SavingsAccount`, deposit, and print kind, holder, and balance. Leave the parent unconstructed.

```
from abc import ABC, abstractmethod


class Account(ABC):
    def __init__(self, holder, balance=0):
        self.holder = holder
        self._balance = float(balance)

    def deposit(self, amount):
```

```
        if amount <= 0:
            print("Invalid amount")
            return
        self._balance += amount

    def get_balance(self):
        return self._balance

    @abstractmethod
    def kind(self):
        pass


class SavingsAccount(Account):
    def kind(self):
        return "savings"


acct = SavingsAccount("Priya", 1000)
acct.deposit(200)
print(acct.kind(), acct.holder, int(acct.get_balance()))
# Account("Campus", 0)  # TypeError — Case-3
```

Expected Output:

```
savings Priya 1200
```

Write a MedCare abstract `Report` with shared `header()` and abstract `title()`. Implement `LabReport` and `InvoiceReport` and print header plus title for each.

```
from abc import ABC, abstractmethod


class Report(ABC):
    def __init__(self, patient):
        self.patient = patient

    def header(self):
        return "MedCare | " + self.patient

    @abstractmethod
    def title(self):
        pass


class LabReport(Report):
    def title(self):
        return "Blood test"


class InvoiceReport(Report):
    def title(self):
        return "Pharmacy bill"


lab = LabReport("Anita")
inv = InvoiceReport("Ravi")
print(lab.header(), "-", lab.title())
print(inv.header(), "-", inv.title())
```

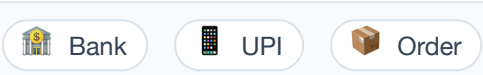
Expected Output:

```
MedCare | Anita – Blood test
MedCare | Ravi – Pharmacy bill
```

---

3. Interface in Python

Real-world analogy: A **wall socket**. Any appliance with the same pins works. The socket does not know *how* the appliance uses power. An **interface** is that socket: only method names (`connect`, `pay`, `send`), no business body. Python has no `interface` keyword — we simulate it with `ABC` + only abstract methods.



Where used? Database drivers (**MySQL** vs **Postgres**), **TechMart** pay methods, **MedCare** SMS vs email.

Definition: If an abstract class contains **only abstract methods**, we consider it an **interface**.

Purpose: Write business code against the contract. Swap the real class later without rewriting the caller.

```
from abc import ABC, abstractmethod


class DBInterface(ABC):
    @abstractmethod
    def connect(self):
        pass

    @abstractmethod
    def disconnect(self):
        pass


class MySQL(DBInterface):
    def connect(self):
        print("Connecting to MySQL...")

    def disconnect(self):
        print("Disconnecting MySQL...")


class Postgres(DBInterface):
    def connect(self):
        print("Connecting to Postgres...")

    def disconnect(self):
        print("Disconnecting Postgres...")


engines = {"MySQL": MySQL, "Postgres": Postgres}
db = engines["MySQL"]()
db.connect()
db.disconnect()
```

Expected Output:

```
Connecting to MySQL...
Disconnecting MySQL...
```

Change the key to `"Postgres"` — the same `connect()` / `disconnect()` calls talk to a different engine.

A class may take **two** interface-style parents: `class WalletAccount(Account, Payable)`. Savings **is an** account. A wallet **can** pay. Do not put `pay` on `Account`, or every savings account is forced to fake a payment.

Interview: `typing.Protocol` describes an interface **without** inheriting `ABC`. Labs use `ABC`.

Remember: Interface = only abstract methods = specification. The child writes the service.

Common Error: storing `holder` / `_balance` on `Payable`. That is account **state**. Keep state on abstract `Account`. Keep `Payable` as methods only.

Note:

- Duck typing (Part 3) fails when we **call** a missing method. ABC fails when we **construct** an incomplete child.
- Wrong: `DBInterface()` — Case-3. Construct `MySQL()` or `Postgres()`.

Write a `Notifier` interface (`ABC` + only `send`). Implement `Sms` and `Email` and loop both to send “Order shipped”.

```
from abc import ABC, abstractmethod


class Notifier(ABC):
    @abstractmethod
    def send(self, msg):
        pass


class Sms(Notifier):
    def send(self, msg):
        print("SMS:", msg)


class Email(Notifier):
    def send(self, msg):
        print("EMAIL:", msg)


for n in (Sms(), Email()):
    n.send("Order shipped")
```

Expected Output:

```
SMS: Order shipped
EMAIL: Order shipped
```

Write `CampusWallet` that implements both `Printable` and `Payable`. Print a slip, pay ₹120, then print the slip again.

```
from abc import ABC, abstractmethod


class Printable(ABC):
    @abstractmethod
    def print_slip(self):
        pass


class Payable(ABC):
    @abstractmethod
    def pay(self, amount):
        pass


class CampusWallet(Printable, Payable):
    def __init__(self, holder, balance):
        self.holder = holder
        self._balance = balance

    def print_slip(self):
        print(self.holder, "₹" + str(int(self._balance)))

    def pay(self, amount):
        if amount <= 0 or amount > self._balance:
            print("pay failed")
            return False
        self._balance -= amount
        print("paid ₹" + str(int(amount)))
        return True


w = CampusWallet("Ravi", 500)
w.print_slip()
w.pay(120)
w.print_slip()
```

Expected Output:

```
Ravi ₹500
paid ₹120
Ravi ₹380
```

---

4. Concrete class vs abstract class vs interface

Real-world analogy: Three documents on a campus project. The **requirement PDF** lists steps only — **interface**. The **half-built floor** has wiring but no shutter — **abstract class**. The **open canteen** is ready to serve — **concrete class**. **SecureBank** `Payable` is the PDF. `Account` is the half-built floor. `SavingsAccount` is the open branch.



Where used? Interview staple: “When interface, when abstract class?” **SecureBank** `Payable` vs `Account`.

1. Know **only** the requirement → **interface**.
2. Some real code, some holes → **abstract class**.

3. Implementation complete → **concrete class**.

	Interface	**Abstract class**	**Concrete class**
What we know	Only method names	Some real code + some holes	Full service
Python	`ABC` + **only** `@abstractmethod`	`ABC` + mix of real methods and abstract methods	Normal class (or child that filled every hole)
Construct?	No	No	Yes
Example	`Payable`, `DBInterface`	`Account` with `deposit` + abstract `kind`	`SavingsAccount`, `MySQL`

The three layers can sit in a chain. Only the bottom class is constructible.

```
from abc import ABC, abstractmethod


class CollegeAutomation(ABC):
    @abstractmethod
    def m1(self):
        pass

    @abstractmethod
    def m2(self):
        pass

    @abstractmethod
    def m3(self):
        pass


class AbsCls(CollegeAutomation):
    def m1(self):
        print("m1 method implementation")

    def m2(self):
        print("m2 method implementation")


class ConcreteCls(AbsCls):
    def m3(self):
        print("m3 method implementation")


c = ConcreteCls()
c.m1()
c.m2()
c.m3()
```

Expected Output:

```
m1 method implementation
m2 method implementation
m3 method implementation
```

AcmeHR onboarding is the same chain: interface lists `register` / `activate`; abstract class writes `register`; intern vs faculty finish `activate`.

Interview: Why not put `pay` on abstract `Account`? Not every account pays. `Payable` is the requirement for types that **can** pay.

Common Error: calling `ABC` "Python's interface keyword." `ABC` is the abstract-class tool. Using it with **only** abstract methods is how we **simulate** an interface.

Keep `pay` off abstract `Account`. Make only `WalletAccount` also `Payable`. Loop accounts and call `pay` only when `isinstance(a, Payable)`.

```
from abc import ABC, abstractmethod


class Account(ABC):
    def __init__(self, holder):
        self.holder = holder

    @abstractmethod
    def kind(self):
        pass


class Payable(ABC):
    @abstractmethod
    def pay(self, amount):
        pass


class SavingsAccount(Account):
    def kind(self):
        return "savings"


class WalletAccount(Account, Payable):
    def kind(self):
        return "wallet"

    def pay(self, amount):
        print("pay ₹" + str(int(amount)), "from", self.holder)
        return True


accounts = [SavingsAccount("Priya"), WalletAccount("Ravi")]
for a in accounts:
    print(a.kind(), a.holder)
    if isinstance(a, Payable):
        a.pay(80)
```

Expected Output:

```
savings Priya
wallet Ravi
pay ₹80 from Ravi
```

Write AcmeHR `Onboarding` (interface), `StaffOnboarding` (implements `register`), and `InternOnboarding` (implements `activate`). Register and activate intern Anita.

```
from abc import ABC, abstractmethod


class Onboarding(ABC):
    @abstractmethod
    def register(self):
        pass

    @abstractmethod
    def activate(self):
        pass


class StaffOnboarding(Onboarding):
    def __init__(self, name):
        self.name = name

    def register(self):
        print("KYC done for", self.name)


class InternOnboarding(StaffOnboarding):
    def activate(self):
        print(self.name, "activated as intern")


intern = InternOnboarding("Anita")
intern.register()
intern.activate()
```

Expected Output:

```
KYC done for Anita
Anita activated as intern
```

Step-by-Step: SecureBank campus wallet

Bank UPI Order

Where used? Same ticket as the lab kit.

`Account` is an **abstract class** (shared deposit / withdraw + hole `kind()`). `Payable` is an **interface** (only `pay`). `SavingsAccount` and `WalletAccount` are **concrete**.

Step 1 — Abstract `Account`

```
from abc import ABC, abstractmethod


class Account(ABC):
    def __init__(self, holder, opening):
        self.holder = holder
        self._balance = float(opening)

    def deposit(self, amount):
        if amount <= 0:
            print("amount must be positive")
            return
        self._balance += amount

    def withdraw(self, amount):
        if amount <= 0:
            print("amount must be positive")
            return
        if amount > self._balance:
            print("insufficient funds")
            return
        self._balance -= amount

    def get_balance(self):
        return self._balance

    @abstractmethod
    def kind(self):
        pass

    def print_slip(self):
        print(self.kind(), "|", self.holder, "₹" + str(int(self._balance)))
```

Step 2 — Interface `Payable` and two concrete children

```
class Payable(ABC):
    @abstractmethod
    def pay(self, amount):
        pass


class SavingsAccount(Account):
    def __init__(self, holder, opening, rate=0.04):
        super().__init__(holder, opening)
        self.rate = rate

    def kind(self):
        return "savings"

    def add_interest(self):
        self.deposit(self.get_balance() * self.rate)


class WalletAccount(Account, Payable):
    def kind(self):
        return "wallet"

    def pay(self, amount):
```

```
        before = self.get_balance()
        self.withdraw(amount)
        if self.get_balance() == before:
            return False
        print("Paid ₹" + str(int(amount)), "from wallet")
        return True
```

Step 3 — One loop

```
accounts = [
    SavingsAccount("Priya", 1000, 0.04),
    WalletAccount("Ravi", 500),
]
for a in accounts:
    a.print_slip()

ok = accounts[1].pay(120)
print("pay ok?", ok)
accounts[1].print_slip()

accounts[0].add_interest()
accounts[0].print_slip()

print("rejected?", WalletAccount("Ravi", 50).pay(120))
```

Expected Output:

```
savings | Priya ₹1000
wallet | Ravi ₹500
Paid ₹120 from wallet
pay ok? True
wallet | Ravi ₹380
savings | Priya ₹1040
insufficient funds
rejected? False
```

Note: Do not write `if a.kind() == "wallet"` to call `pay`. Use `isinstance(a, Payable)`. `print_slip` is a normal method on `Account` — that is why `Account` is not an interface.

Practice Exercises

Basic

1. What is an abstract method in one or two sentences?

   Model answer:

   A method we declare even though we do not know the implementation yet. In Python: `@abstractmethod` with `pass`. Child classes write the real body.

2. Why do we import from `abc`? What error do we get if we skip it?

   Model answer:

   `@abstractmethod` and `ABC` live in the `abc` module. Skip the import → `NameError: name 'abstractmethod' is not defined`.

3. Write the conclusion of Cases 1–5 in one line.
   Model answer:
   If a class contains at least one abstract method **and** extends `ABC`, instantiation is not possible.

Intermediate

1. `class Bus(Vehicle): pass` when `Vehicle` is `ABC` with abstract `no_of_wheels`. Is the `class` line legal? Is `Bus()` legal?
   Model answer:
   The class line is legal. `Bus()` is not — `Bus` did not override `no_of_wheels`, so it is still abstract (`TypeError`).
2. Create abstract `Shape` with `area()`, `Circle(r)` using `3.14 * r * r`, `Rect(w, h)` as `w * h`. Print both areas as integers.
   Model answer:

```
from abc import ABC, abstractmethod


class Shape(ABC):
    @abstractmethod
    def area(self):
        pass


class Circle(Shape):
    def __init__(self, r):
        self.r = r

    def area(self):
        return 3.14 * self.r * self.r


class Rect(Shape):
    def __init__(self, w, h):
        self.w = w
        self.h = h

    def area(self):
        return self.w * self.h


print(int(Circle(2).area()))
print(int(Rect(3, 4).area()))
```

Expected Output:

```
12
12
```

1. When do we pick interface vs abstract class vs concrete class?
   Model answer:
   Interface: we know only the requirement (`connect` / `pay`). Abstract class: we have some real code and some holes (`deposit` written, `kind` missing). Concrete class: implementation is complete and we can create objects (`MySQL()`, `SavingsAccount(...)`).

Mini-Project (Assignment): SecureBank Campus Wallet

Lab kit: 15_Lab_Assets/32_campus_wallet_abc/

Business request: SecureBank campus desk needs savings and wallet on one slip loop. Wallet can **pay**. A generic `Account(...)` must not be allowed.

What to build

Piece	Detail
`Account(ABC)`	abstract class: holder, balance, `deposit` / `withdraw` / `get_balance` / `print_slip`; abstract `kind()`
`Payable`	interface: only `pay(amount)`
`SavingsAccount`	concrete; `kind()` → `"savings"`; `add_interest()`
`WalletAccount`	concrete; `Account` **and** `Payable`; `pay` uses `withdraw` (print + return)
Demo	list of accounts; print slips; pay from wallet; add interest on savings

Stretch

- Reject a zero / negative pay with a print, keep the old balance.
- Add a `CurrentAccount` that is an `Account` but **not** `Payable`.

Done when

- [ ] `Account("Campus", 0)` is not allowed
- [ ] Savings and wallet both print a slip through the parent type
- [ ] Wallet `pay` succeeds and fails (print + return) without `raise`
- [ ] Savings can `add_interest()`; it does not implement `pay`

Interview Q&A

Shared with students after concepts and the mini-project. Use for review and interview prep — questions and model answers are included.

Q1. What is an abstract method?
A: A method with declaration only, no real implementation. Python: `@abstractmethod` and `pass`. Child classes write the body.

Q2. What is an abstract class? Can we instantiate it?
A: A partially implemented class. Extend `ABC` and include at least one abstract method. Then we cannot create an object (`TypeError`). Construct the concrete child instead.

Q3. Extending `ABC` with no abstract method — can we create an object?
A: Yes (Case-2). Both `ABC` **and** an abstract method are required to block instantiation.

Q4. Child does not override the abstract method. What happens?
A: The child is also abstract. The `class` line is fine. Creating the child object raises `TypeError`.

Q5. What is an interface in Python?

A: There is no `interface` keyword. An abstract class with **only** abstract methods is the interface. Example: `DBInterface` with `connect` / `disconnect`; `MySQL` and `Postgres` implement it.

Q6. When interface, when abstract class, when concrete class?

A: Interface = we know only the requirement. Abstract class = partial implementation. Concrete class = complete implementation, ready to create objects.

Q7. Why not put `pay` on `Account`?

A: Not every account pays a merchant. `Payable` is a separate requirement. Wallet opts in; savings does not fake `pay`.

Q8. Can an abstract class contain normal methods?

A: Yes. That is the difference from an interface. `deposit` / `print_slip` on `Account` are normal; only `kind()` is abstract.

Q9. `ABC` vs `Protocol`?

A: `ABC` — inherit the contract; missing methods fail at construction. `Protocol` — structural ("if it has `pay`, it counts"); mainly for type checkers. Labs use `ABC`.

Q10. Why give an abstract class `__init__`?

A: Children still need holder and opening balance. `__init__` runs through `super()` when we construct the child — never as `Account(...)`.

Q11. Duck typing vs ABC — what extra do we get from `@abstractmethod`?

A: Duck typing fails when we **call** a missing method. An ABC fails when we **construct** an incomplete child. Named contracts also let us write `isinstance(x, Payable)`.

Next

Finish the Campus Wallet lab kit, then continue with **Exception Handling**.